
MENTAL HEALTH DETECTOR

*AN AI-POWERED TOOL FOR DETECTION
AND TREATMENT OF MENTAL HEALTH
DISORDERS.*

BY GROUP 2.

INTRODUCTION.

- This project is an integrated mental health assessment tool combining:
 - ❑ **Signs-Based Disorder Prediction** – Uses NLP to analyse symptom descriptions, identify likely mental disorders, and provide tailored self-care and professional recommendations.
 - ❑ **Lifestyle-Based Risk Assessment** – Uses machine learning on structured lifestyle and demographic data to predict mental health risk and severity.
- Together, these features enable holistic evaluation, early detection, risk scoring, and actionable advice for individuals and practitioners, promoting awareness and informed decisions.



PROBLEM STATEMENT.



Background

Mental health affects daily life, but stigma, cost, and limited access—especially in low- and middle-income regions—create barriers to care. With over 1 in 8 people worldwide affected, AI offers a way to detect emotional distress through text, provide supportive feedback, and assist therapists by tracking emotional trends, improving both individual and professional care.



Problem Statement

Many face untreated mental health challenges due to cost, stigma, and lack of access. Emotional distress is often untracked, and therapists have limited insight between sessions. There's a need for private, accessible tools that enable self-reflection and offer valuable data to both individuals and mental health professionals.

OBJECTIVES.



Analysis-Based Objectives

Detect emotional and linguistic patterns in user text linked to conditions like anxiety, depression, and burnout.

Identify symptom trends and frequencies signaling onset or escalation of challenges.

Classify emotional expressions into broad mental health categories using historical data.



Feature Engineering Objectives

Preprocess text (tokenization, lemmatization, stop-word removal) for clean model input.

Extract features such as emotional tone, symptom keywords, frequency/persistence of states, and time-based changes.



Modelling Objectives

Build and fine-tune a multi-label NLP model to detect multiple co-occurring conditions.

Integrate into an empathetic chatbot that offers daily check-ins, coping strategies, and motivational content.

Assess distress levels to tailor tone and urgency.

DATA DESCRIPTION AND PREPARATION

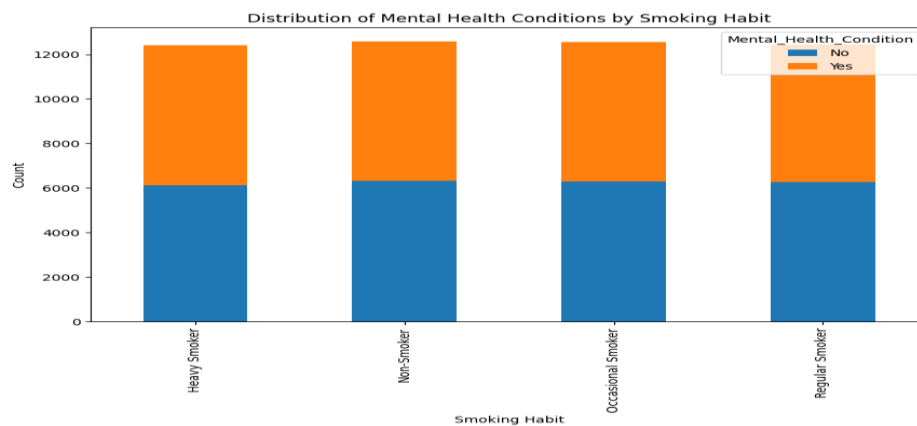
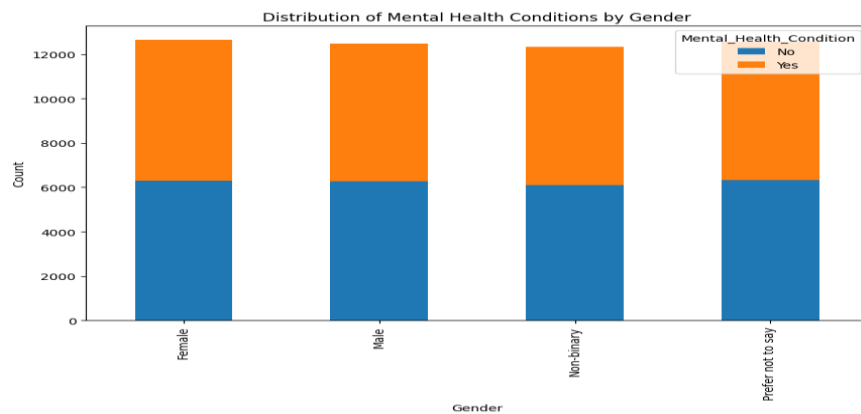
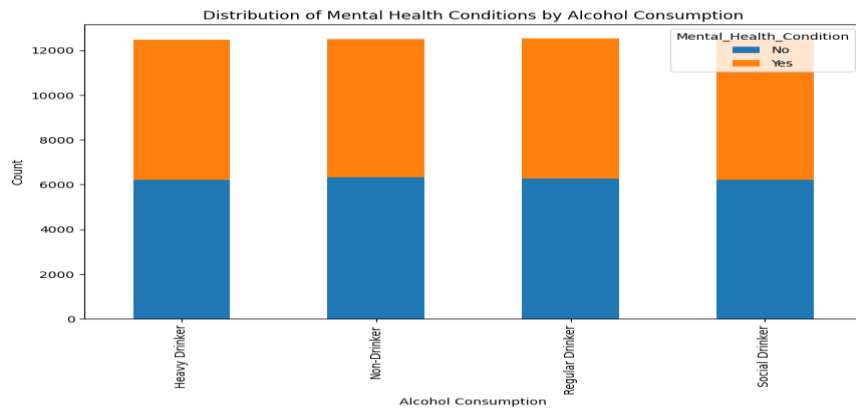
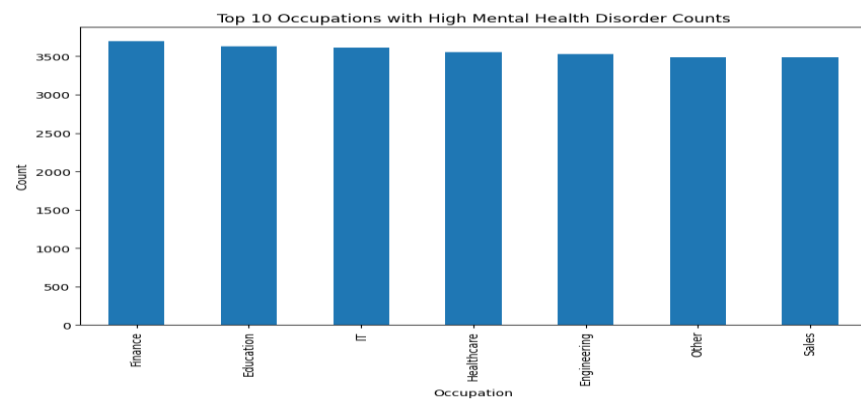
1. Mental Health Content (mentalhealth.org.uk)

- **Content:** Signs, symptoms, emotional indicators, coping strategies, recommendations of 19 disorders ; (Anxiety, Attention Deficit Hyperactivity Disorder (ADHD), Autism, Bipolar disorder, Loneliness, Dementia, Depression, Menopause, OCD, Panic Attacks, Personality Disorder, PTSD, Postnatal-Depression, Psychosis, Schizophrenia, Self - Harm, Stress, Suicidal Thoughts, Trauma)
- **Purpose:** Train emotional tone detection & generate chatbot responses.
- **Processing:**
 - Clean text data, no missing values.
 - SentenceBERT embeddings for classification & response modeling.

2. Demographic & Behavioral Data

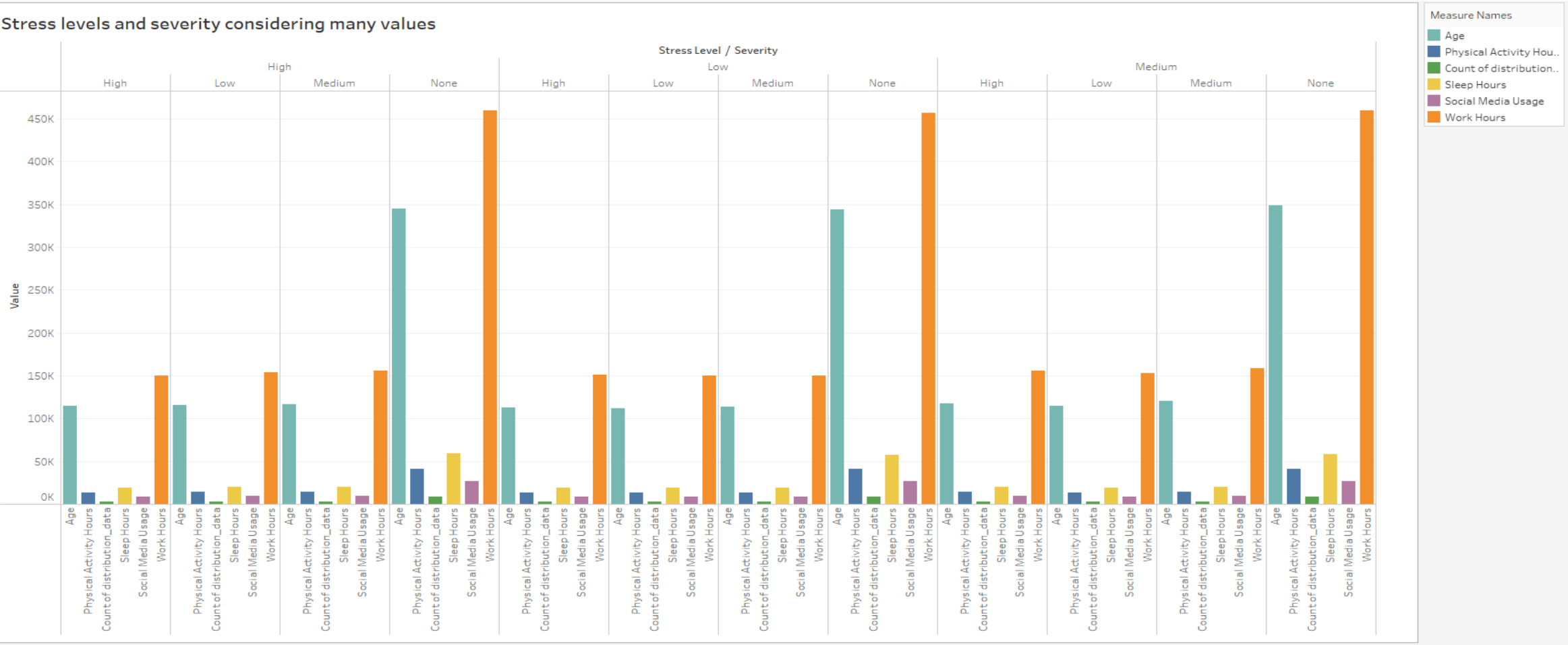
- **Features:** Age, gender, occupation, diet quality, smoking, alcohol use, sleep hours, physical activity, stress level, social media use, consultation history, medication, work hours.
- **Purpose:** Train CatBoost model for multi-label classification & risk prediction.
- **Preparation:**
 - Handled missing severity values via mode imputation.
 - Normalized and encoded categorical variables.

EXPLORATORY DATA ANALYSIS.



EXPLORATORY DATA ANALYSIS

Stress levels and severity considering many values



MODELLING AND EVALUATION.



1. Text-Based Modeling

- **Input:** User symptom descriptions & journal entries
- **Processing:** SentenceBERT embeddings → Multi-label classifier
- **Goal:** Detect conditions like depression, anxiety, burnout
- **Integration:** Empathetic chatbot offering condition-specific suggestion.

2. Structured Data Modeling

- **Input:** 13 behavioral & demographic features (age, gender, occupation, alcohol, smoking, sleep, physical activity, stress, etc.)
- **Models:** Stacked Gradient Boosting + CatBoost (also LightGBM in ensemble)
- **Target:** Multi-label classification + severity prediction (mild, moderate, severe)
- **Evaluation Metrics.**
 - Average Severity Precision; 0.667,
 - Average Severity Recall; 1

DEPLOYMENT

- **Platform:**
 - **Streamlit:** Local interface for running the app
 - **FastAPI:** Backend for scalable model inference
 - **Ngrok:** Expose app online for testing
- **Modes:**
 - **Signs-Based:** NLP prediction from user-described symptoms
 - **Lifestyle-Based:** Structured data prediction from age, stress, sleep, etc.

The screenshot shows the 'Signs-Based Prediction' tab selected. It features a text input field for entering emotional signs or feelings, with a placeholder example: 'e.g., I feel exhausted, anxious, and hopeless'. Below the input field is a green button labeled 'Predict Disorder'.

The screenshot shows the 'Lifestyle Risk & Severity Estimation' interface. It contains several input fields and sliders for structured data:

- Age:** Slider set to 25.
- Sleep Hours:** Slider set to 6.
- Gender:** Dropdown menu set to 'Female'.
- Work Hours/week:** Slider set to 40.
- Occupation:** Dropdown menu set to 'Education'.
- Physical Activity (hrs/week):** Slider set to 3.
- Consultation History:** Dropdown menu set to 'No'.
- Social Media (hrs/day):** Slider set to 3.00.
- Stress Level:** Dropdown menu set to 'High'.
- Diet Quality:** Dropdown menu set to 'Average'.
- Smoking Habit:** Dropdown menu (partially visible).

CONCLUSIONS AND RECOMMENDATIONS

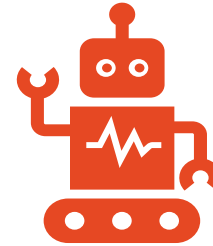


Conclusions.

The Mental Health Prediction Assistant combines NLP-based symptom analysis with structured lifestyle data for early detection and personalized support.

It provides scalable mental health screening, tailored recommendations, and a modular, user-friendly architecture suitable for digital health platforms or personal self-assessment.

Overall, it helps bridge the gap between early signs of distress and timely intervention.



Top Recommendations

Model Performance Improvement: Use advanced NLP embeddings (e.g., OpenAI embeddings, ClinicalBERT) to enhance classifier accuracy.

User-Centric Refinement: Integrate user feedback into the UI to dynamically improve predictions and recommendations.

Integration of Multi-Source Data: Incorporate wearable health tracker data (sleep, heart rate, activity) to improve prediction reliability.

NEXT STEPS



Continuous Learning System - Build a feedback loop where user-confirmed diagnoses and lifestyle updates are fed back into the system for continuous model retraining.



Multilingual Support - Expand the system to handle multiple languages to serve a wider audience.



Regulatory Compliance - Adapt the platform to comply with HIPAA, GDPR, and other health data regulations depending on target regions for deployment.

**THANK
YOU!**

